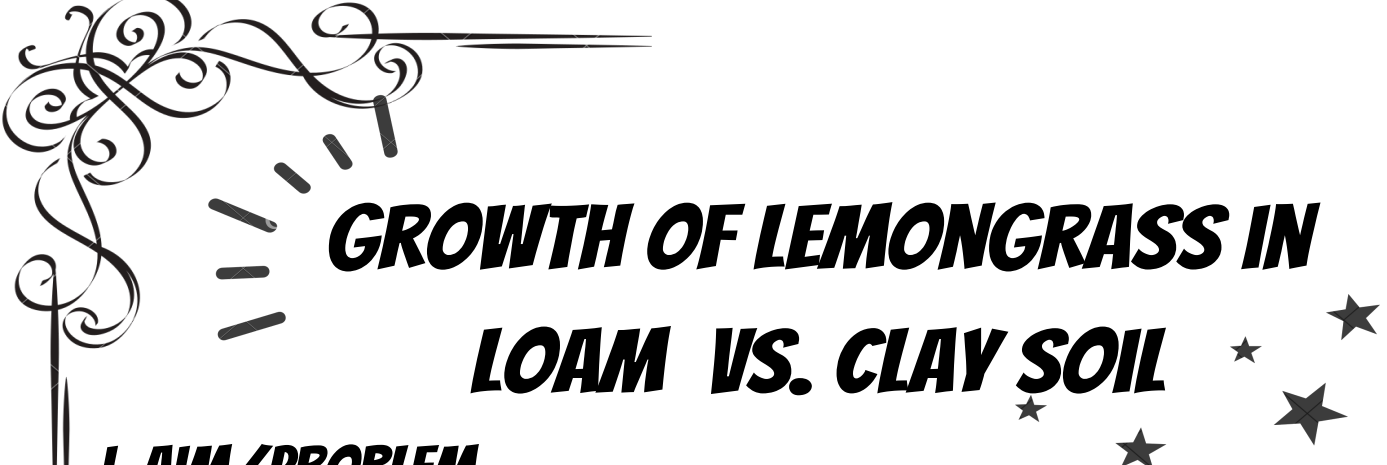


# Asexual Reproduction

SCIENTIFIC INVESTIGATION





# ***GROWTH OF LEMONGRASS IN LOAM VS. CLAY SOIL***

## ***I. AIM/PROBLEM***

- Which soil type produces faster and healthier lemongrass growth over a period of 3 weeks?

### **hypothesis**

- If lemongrass are grown in a loam soil, then it will grow taller and healthier.

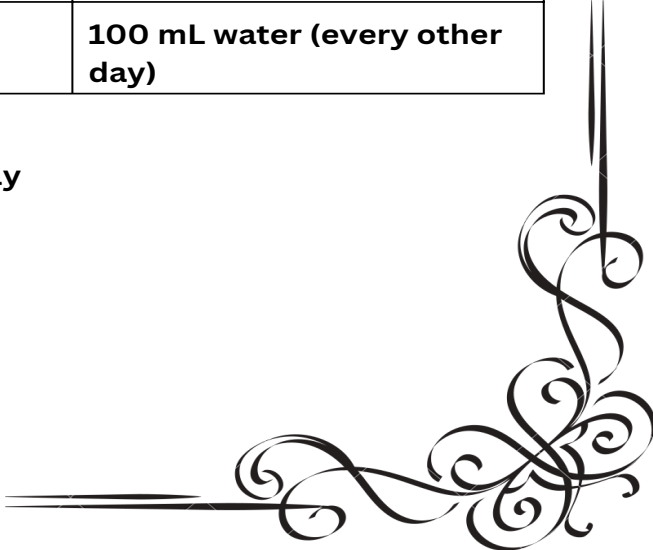
## ***II. EXPERIMENT***

### **Materials**

- 2 identical cups with drainage holes
- 2 healthy lemongrass cuttings
- 200g of loam soil
- 200g of clay soil
- meter stick
- 100 ml of water

| Week   | Day/s     | Materials Used                                                                       |
|--------|-----------|--------------------------------------------------------------------------------------|
| Week 1 | Day 1     | 2 small plastic cups, 200g garden soil, 200g clay soil, 2 lemongrass cuttings, 100ml |
|        | Day 2–7   | 100 mL water (every other day)                                                       |
| Week 2 | Day 8–14  | 100mL water (every other day),                                                       |
| Week 3 | Day 15–21 | 100 mL water (every other day)                                                       |

### **Procedure**

- Get two cups and put the loam soil in one cup and clay soil in other cup.
  - Plant one lemongrass cutting in each pot.
  - Water both pots equally every other day.
  - Place them in the same spot with sunlight.
  - Observe their growth for four weeks.
  - Measure and record their height each week.
  - Compare which one grows better.
- 

## DATA AND OBSERVATION

Data:

**TABLE 1**

| NAME OF THE PLANT<br>(scientific name) | NUMBER OF INDIVIDUALS INVOLVED | TYPE OF ASEXUAL REPRODUCTION | PLANT PART USED IN REPRODUCTION |
|----------------------------------------|--------------------------------|------------------------------|---------------------------------|
| Lemongrass<br>(Cymbopogon citratus)    | 2                              | vegetative reproduction      | stem or stalk                   |

Observation:

**TABLE 2**

| DAY          |                     | Observation                                                                         |        |           |
|--------------|---------------------|-------------------------------------------------------------------------------------|--------|-----------|
| 7            |                     | the lemongrasses are not growing yet                                                |        |           |
| 14           |                     | The lemongrass in the loam soil are more longer and higher than clay soil           |        |           |
| 21           |                     | The lemongrass in clay soil are bigger, longer and higher than the one in loam soil |        |           |
| Type of Soil | Initial Height (cm) | Week 1 cm                                                                           | Week 2 | Week 3 cm |
| Garden Soil  | 10                  | 8                                                                                   | 16     | 12        |
| Clay Soil    | 10                  | 12                                                                                  | 15     | 24        |

## Guide Questions:

1.What is the independent variable in your experiment

- Type of soil used (loam soil and clay soil)

2.In which set up did your plant grow longer/better?

- The plant grew better in clay soil.

3.If you will repeat the experiment, what changes will you make to grow the plant better/longer?

- If I will repeat the experiment, I will add more water and place the plant in a sunnier area and plant it in a clay soil so it can grow better and longer.

## Conclusion

- Therefore, the tanglad grew better in clay soil because it was able to hold more moisture, which helped the plant stay healthy and grow longer.

## ***LEMONGRASS FROM DAY 1-21***

Day 1



Day 2



Day 3



Day 4



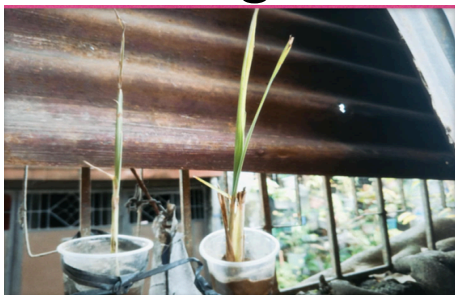
Day 5



Day 6



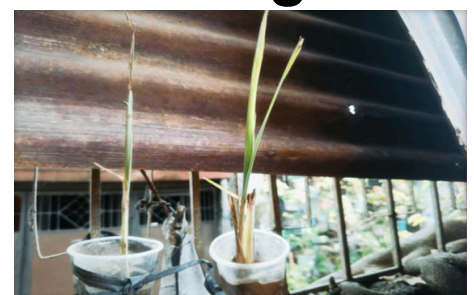
Day 7



Day 8



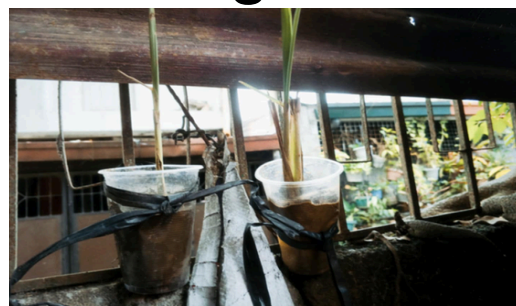
Day 9



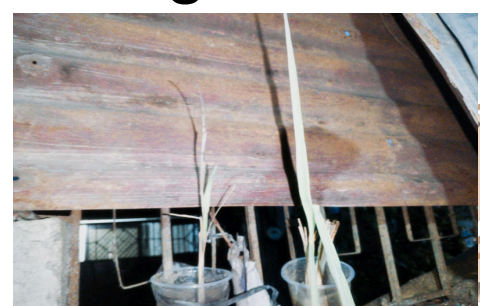
Day 10



Day 11



Day 12-13





**Day 13-16**



**Day 16-18**



**Day 18-20**



**Day 21**



## ***INDEPENDENT VARIABLE***

- Type of soil (garden soil and clay soil)

## ***DEPENDENT VARIABLE***

- Growth of tanglad (height and health of the plant)

## ***CONTROLLED VARIABLE***

- Amount of water, amount of sunlight, size of container, and number of tanglad plants used

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